AI-Driven Resource Demand- and Allocation

Recommendation for CAPEX Portfolio Management

Master's Thesis

submitted by

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Declaration of Authenticity

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Abstract

Managing large-scale Capital Expenditure (CAPEX) portfolios requires the coordinated execution of numerous parallel projects with complex interdependency structures. A fundamental challenge in this context is the lack of transparency concerning specific skill requirements and their temporal distribution across the project lifecycle. Without precise, forward-looking insights into which competencies are required at which point in time, organizations are unable to proactively ensure on-time project delivery. Existing manual planning methods frequently fail to identify critical resource bottlenecks sufficiently early, resulting in project delays, cost overruns, and suboptimal investment outcomes.

This thesis develops a generalizable, application-oriented framework for an AI-supported recommendation engine designed to enhance strategic resource management in CAPEX portfolio environments. The framework addresses three interrelated challenges: (1) *temporal skill-demand forecasting*, which utilizes advanced analytical models to predict the specific skill sets and capacities required across different project phases; (2) *proactive bottleneck identification*, which detects potential resource scarcities early enough to enable timely strategic adjustments and risk mitigation; and (3) *optimized allocation recommendations*, which suggest the most efficient distribution of available resources to meet project timelines while respecting organizational constraints.

The proposed system integrates machine learning techniques for demand forecasting with optimization algorithms for allocation planning, embedded in a decision-support architecture. Evaluation results demonstrate that ...

Keywords: CAPEX portfolio management, resource allocation, skill-demand forecasting, bottleneck detection, AI recommendation engine, optimization

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List of Abbreviations

AI Artificial Intelligence

CAPEX Capital Expenditure

OPEX Operational Expenditure

ML Machine Learning

DL Deep Learning

NLP Natural Language Processing

PPM Project Portfolio Management

ERP Enterprise Resource Planning

KPI Key Performance Indicator

FTE Full-Time Equivalent

LSTM Long Short-Term Memory

RF Random Forest

XGBoost Extreme Gradient Boosting

API Application Programming Interface

ROI Return on Investment

WBS Work Breakdown Structure

DSS Decision Support System

LP Linear Programming

MIP Mixed-Integer Programming

Chapter 1

Introduction

1.1 Motivation and Problem Statement

Capital Expenditure (CAPEX) portfolios in large organizations encompass dozens to hundreds of concurrent investment projects, each competing for a shared and finite pool of specialized resources. Unlike operational tasks, CAPEX projects — infrastructure upgrades, plant expansions, technology rollouts, or engineering programs — are characterized by long planning horizons, significant upfront commitments, and complex interdependencies between workstreams. A delay in one project can cascade across the entire portfolio, amplifying costs and disrupting strategic timelines.

At the core of effective CAPEX portfolio execution lies the question of *resource allocation*: which personnel, with which skills, are needed where, and when? This seemingly operational question has profound strategic consequences. Skills in areas such as civil engineering, electrical systems integration, or project controls are neither fungible nor immediately scalable. When demand for a specific competency peaks simultaneously across multiple projects — a phenomenon known as a *resource bottleneck* — the resulting contention can stall progress on high-priority investments.

Despite the magnitude of these risks, most organizations continue to rely on manual, spreadsheet-driven planning processes that lack the analytical depth to anticipate bottlenecks before they materialize. Current tools are largely reactive: they report shortages after scheduling conflicts have already occurred rather than forecasting them weeks or months in advance when corrective actions — such as contractor procurement, internal transfers, or scope adjustments — are still viable.

This thesis addresses this gap by developing an AI-driven recommendation engine for resource demand forecasting and allocation optimization in CAPEX portfolio environments.

1.2 Research Objectives

This research pursues three interrelated objectives:

1. **Temporal Skill-Demand Forecasting:** Develop models that predict the type, volume, and timing of skill requirements across project phases within a CAPEX portfolio.

2. **Proactive Bottleneck Identification:** Design algorithms that detect resource scarcities at sufficient lead time to enable proactive mitigation strategies.

3. **Optimized Allocation Recommendations:** Formulate and solve resource allocation problems that suggest the most efficient distribution of available resources while respecting organizational and project constraints.

1.3 Thesis Statement

This thesis argues that an AI-driven recommendation engine — integrating temporal skill-demand forecasting, proactive bottleneck detection, and constraint-aware allocation optimization — can measurably improve resource planning outcomes in CAPEX portfolio management compared to conventional manual approaches. Specifically, the claim is that such a system enables earlier identification of resource scarcities and more efficient utilization of available competencies, thereby reducing project delays attributable to resource contention.

1.4 Research Questions

The following research questions guide this work:

RQ1 How can historical project data be leveraged to accurately forecast skill-specific resource demand across the lifecycle phases of CAPEX projects?

RQ2 Which machine learning architectures are most suitable for temporal resource demand forecasting in the context of heterogeneous CAPEX portfolios?

RQ3 How can detected demand forecasts be translated into actionable, constraint-aware allocation recommendations that support portfolio managers?

RQ4 To what extent does an AI-driven recommendation approach outperform conventional manual planning methods in terms of bottleneck anticipation and resource utilization?

1.5 Scope and Delimitations

This thesis focuses on the strategic and tactical planning layers of resource management — specifically demand forecasting and allocation recommendation — rather than on real-time operational scheduling. The framework is designed to be organization-agnostic and generalizable across industry sectors where CAPEX portfolio management is practiced, though empirical evaluation is conducted in the context of ...

The following aspects are explicitly outside the scope of this thesis:

- Individual project scheduling at the task level (Gantt-level granularity)
- Financial capital allocation or CAPEX budget optimization
- HR performance management or workforce development planning

1.6 Thesis Structure

The remainder of this thesis is organized as follows:

Chapter 2 establishes the theoretical and conceptual foundations of CAPEX portfolio management, resource planning, and AI-based forecasting and optimization.

Chapter 3 surveys existing literature and tools in the areas of project resource management, demand forecasting, and AI-based decision support systems.

Chapter 4 describes the research design, data sources, modeling approach, and evaluation strategy.

Chapter 5 presents the conceptual architecture of the proposed AI recommendation engine.

Chapter 6 details the technical implementation of the framework components.

Chapter 7 reports the empirical evaluation results and benchmarks against baseline methods.

Chapter 8 interprets the findings, discusses limitations, and reflects on practical implications.

Chapter 9 summarizes the contributions of this thesis and outlines directions for future research.

Chapter 2

Theoretical Background

2.1 CAPEX Portfolio Management

2.1.1 Definition and Characteristics

2.1.2 Project Lifecycle Phases in CAPEX Contexts

2.1.3 Interdependency Structures in Large Portfolios

2.2 Resource Management in Project Portfolios

2.2.1 Skill-Based Resource Planning

2.2.2 Resource Bottlenecks: Definition and Impact

2.2.3 Current Planning Practices and Their Limitations

2.3 Forecasting Methods

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2.4.2 Mathematical Programming Approaches

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2.5 The AI-4-SME Framework

The AI-4-SME Framework (Peter et al., 2025), developed at FHNW School of Business

Design, Build, Run — for identifying and implementing AI solutions in enterprise contexts. It is adopted as the overarching methodological structure of this thesis.

2.5.1 Design Phase: KIAs and DIAs

A central concept of the framework is the distinction between *Knowledge-Intensive Activities* (KIAs) and *Data-Intensive Activities* (DIAs). KIAs require cognitive capabilities such as problem solving, analysis, creativity and decision making; DIAs primarily involve data analysis and transaction processing. In the CAPEX resource management context, skill-demand forecasting and bottleneck detection are KIAs, while allocation computation is a DIA.

2.5.2 Build Phase: Hybrid AI Process Model

The framework defines two development paths — a data-based path (ML with TensorFlow/PyTorch) and a knowledge-based path (knowledge graphs with Neo4j, GraphDB, Stardog, RDFox, metaphactory, or TopBraid Composer) — which can be combined into a hybrid AI solution. The interaction between both paths is a central feature of the Build phase.

2.5.3 Run Phase: Governance and Integration

Deployment addresses corporate governance (GDPR, EU AI Act), IT infrastructure decisions, API integration with ERP/CRM systems, and change management for human-AI collaboration.

2.6 AI-Based Decision Support Systems

2.6.1 Recommendation Engines

2.6.2 Explainability and Trust in AI Recommendations

Chapter 3

Related Work

3.1 Resource Management in Project Portfolio Management

3.2 Machine Learning for Project Resource Forecasting

3.3 Optimization-Based Resource Allocation

3.4 AI-Assisted Decision Support in Enterprise Contexts

3.5 Gap Analysis

Chapter 4

Research Methodology

4.1 Research Design

This thesis follows a design science research (DSR) paradigm (Hevner et al., 2004), which is well-suited for the development and evaluation of IT artifacts in applied organizational contexts. DSR emphasizes the creation of purposeful artifacts — in this case, a forecasting and allocation recommendation framework — and their rigorous evaluation against real-world problems.

4.2 Data Sources and Data Preparation

4.2.1 Data Sources

4.2.2 Data Preprocessing

4.2.3 Data Quality and Limitations

4.3 Modeling Approach

4.3.1 Demand Forecasting Model

4.3.2 Bottleneck Detection Algorithm

4.3.3 Allocation Optimization Model

Let the set of projects be $\mathcal{P} = \{p_1, p_2, \ldots, p_n\}$, the set of skill categories $\mathcal{S} = \{s_1, s_2, \ldots, s_m\}$, and the planning horizon $\mathcal{T} = \{t_1, t_2, \ldots, t_T\}$.
For each project $p \in \mathcal{P}$, skill $s \in \mathcal{S}$, and time period $t \in \mathcal{T}$, let:

- $d_{p,s,t}$ — forecasted demand for skill s in project p at time t (in FTE)

- $c_{s,t}$ — available capacity of skill s at time t (in FTE)
- $x_{p,s,t}$ — allocated resource of skill s to project p at time t (decision variable)

The optimization objective minimizes the total unmet demand:

$$\min \sum_{p \in \mathcal{P}} \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}} \max(0,\ d_{p,s,t} - x_{p,s,t}) \tag{4.1}$$

Subject to:

$$\sum_{p \in \mathcal{P}} x_{p,s,t} \leq c_{s,t} \qquad \forall s \in \mathcal{S},\ t \in \mathcal{T} \tag{4.2}$$

$$x_{p,s,t} \geq 0 \qquad \forall p, s, t \tag{4.3}$$

4.4 Evaluation Strategy

4.4.1 Forecasting Accuracy Metrics

4.4.2 Bottleneck Detection Performance

4.4.3 Allocation Quality Metrics

4.4.4 Baseline Comparisons

Chapter 5

System Design

5.1 Architecture Overview

[Architecture Diagram — TODO]

Figure 5.1: High-level architecture of the AI-driven resource recommendation engine.

5.2 Data Ingestion and Integration Layer

5.3 Demand Forecasting Module

5.4 Bottleneck Detection Module

5.5 Allocation Recommendation Module

5.6 Recommendation Interface

Chapter 6

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Chapter 8

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8.1 Interpretation of Results

8.2 Generalizability

8.3 Limitations

8.4 Practical Implications

Chapter 9

Conclusion

9.1 Summary of Contributions

This thesis developed an AI-driven framework for resource demand forecasting and allocation recommendation in CAPEX portfolio management. The principal contributions are:

1. A temporal skill-demand forecasting model trained on historical CAPEX project data, capable of predicting skill-specific resource requirements across project lifecycle phases.
2. A proactive bottleneck detection algorithm that identifies anticipated resource scarcities at sufficient lead time for corrective action.
3. A constraint-aware optimization model that translates demand forecasts into actionable allocation recommendations aligned with organizational constraints.
4. An integrated system architecture embedding the above components into a cohesive decision-support engine.

9.2 Answers to Research Questions

9.3 Future Work

Several avenues remain open for future research:

- **Real-time adaptation:** Extending the framework to incorporate live project status updates and dynamically re-optimize allocations.
- **Multi-objective optimization:** Incorporating additional objectives such as cost minimization, employee workload balancing, or skill development goals.

- **Cross-portfolio learning:** Training models on data from multiple organizations to improve generalizability and reduce data requirements per site.
- **Natural language interfaces:** Enabling portfolio managers to query the system and receive recommendations through conversational AI interfaces.

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Peter, M. K., Laurenzi, E., and Hinkelmann, K. (2025). *Workshop Canvas AI-4-SME Framework: Identification and utilisation of strategic AI opportunities to strengthen a firm's competitiveness and innovation capabilities*. FHNW School of Business, Switzerland. Version 06/25. Available at https://www.ai-framework.net.

Appendix A

Supplementary Material

A.1 Feature Descriptions

A.2 Hyperparameter Configurations

A.3 Additional Evaluation Plots